

Portfolio Analysis

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This report uses Monte Carlo simulation to compare the 10 day Value at Risk and Conditional Value at Risk of a diversified equity portfolio against a concentrated technology portfolio, both equally weighted across 11 large cap stocks. The diversified portfolio's simulated risk was consistently about half that of the concentrated portfolio, a gap that is consistent with a lower average correlation among its holdings. Results were validated through historical backtesting. Both portfolios' return distributions were found to be leptokurtic, which is a source of tail risk that is not captured by the model's assumption of normality.

```
tickers_A <- c("AAPL", "MSFT", "XOM", "CVX", "JNJ", "PFE", "JPM", "BAC", "PG", "KO", "CAT")
tickers_B <- c("AAPL", "MSFT", "NVDA", "GOOGL", "META", "AMZN", "AVGO", "CRM", "ADBE", "ORCL")
start_date <- "2021-01-01"
end_date <- "2026-09-07"
portfolio_A <- tq_get(tickers_A,
  from = start_date,
  to = end_date,
  get = "stock.prices")
portfolio_B <- tq_get(tickers_B,
  from = start_date,
  to = end_date,
  get = "stock.prices")
portfolio_A |>
  group_by(symbol) |>
  summarise(n = n(), n_na = sum(is.na(adjusted)))
```

```
# A tibble: 11 x 3
  symbol      n  n_na
  <chr>   <int> <int>
1 AAPL   1425     0
2 BAC    1425     0
3 CAT    1425     0
4 CVX    1425     0
```

5	JNJ	1425	0
6	JPM	1425	0
7	KO	1425	0
8	MSFT	1425	0
9	PFE	1425	0
10	PG	1425	0
11	XOM	1425	0

```
portfolio_B |>
  group_by(symbol) |>
  summarise(n = n(), n_na = sum(is.na(adjusted)))
```

```
# A tibble: 11 x 3
  symbol      n  n_na
  <chr> <int> <int>
1 AAPL   1425     0
2 ADBE   1425     0
3 AMD    1425     0
4 AMZN   1425     0
5 AVGO   1425     0
6 CRM    1425     0
7 GOOGL  1425     0
8 META   1425     0
9 MSFT   1425     0
10 NVDA   1425     0
11 ORCL   1425     0
```

```
portfolio_A <- portfolio_A |>
  group_by(symbol) |>
  arrange(date) |>
  mutate(daily_return = log(adjusted / lag(adjusted))) |>
  ungroup()
```

```
portfolio_B <- portfolio_B |>
  group_by(symbol) |>
  arrange(date) |>
  mutate(daily_return = log(adjusted / lag(adjusted))) |>
  ungroup()
```

```
portfolio_A |>
  group_by(symbol) |>
```

```
summarise(mean_return=mean(daily_return, na.rm = T),
          sd_return=sd(daily_return, na.rm = T),
          .groups = "drop")
```

```
# A tibble: 11 x 3
  symbol mean_return sd_return
  <chr>      <dbl>      <dbl>
1 AAPL      0.000657      0.0175
2 BAC        0.000613      0.0167
3 CAT        0.00112      0.0195
4 CVX        0.000801      0.0160
5 JNJ        0.000507      0.0109
6 JPM        0.000836      0.0152
7 KO         0.000475      0.0103
8 MSFT       0.000617      0.0172
9 PFE        0.0000257     0.0155
10 PG        0.000146      0.0112
11 XOM       0.00110      0.0172
```

```
portfolio_B |>
  group_by(symbol) |>
  summarise(mean_return=mean(daily_return, na.rm = T),
            sd_return=sd(daily_return, na.rm = T),
            .groups = "drop")
```

```
# A tibble: 11 x 3
  symbol mean_return sd_return
  <chr>      <dbl>      <dbl>
1 AAPL      0.000657      0.0175
2 ADBE     -0.000421      0.0232
3 AMD        0.00115      0.0346
4 AMZN       0.000340      0.0221
5 AVGO       0.00157      0.0267
6 CRM        0.000126      0.0238
7 GOOGL      0.000966      0.0197
8 META       0.000589      0.0275
9 MSFT       0.000617      0.0172
10 NVDA      0.00202      0.0318
11 ORCL      0.000696      0.0258
```

```
returns_wide_A <- portfolio_A |>
  filter(!is.na(daily_return)) |>
  select(date, symbol, daily_return) |>
  pivot_wider(names_from = symbol, values_from = daily_return)
```

```
returns_wide_B <- portfolio_B |>
  filter(!is.na(daily_return)) |>
  select(date, symbol, daily_return) |>
  pivot_wider(names_from = symbol, values_from = daily_return)
```

```
cor_matrix_A <- cor(returns_wide_A |>
  select(-date),
  use = "pairwise.complete.obs")
cor_matrix_B <- cor(returns_wide_B |>
  select(-date),
  use = "pairwise.complete.obs")
cor_matrix_A
```

	AAPL	MSFT	XOM	CVX	JNJ	PFE	JPM
AAPL	1.0000000	0.54429783	0.15181817	0.18902116	0.14432504	0.1979821	0.3269465
MSFT	0.5442978	1.0000000	0.04544604	0.10719657	0.01264437	0.1263269	0.2700462
XOM	0.1518182	0.04544604	1.0000000	0.85535759	0.13064261	0.1212986	0.3055003
CVX	0.1890212	0.10719657	0.85535759	1.0000000	0.15002820	0.1590103	0.3450506
JNJ	0.1443250	0.01264437	0.13064261	0.15002820	1.0000000	0.4109199	0.1784448
PFE	0.1979821	0.12632694	0.12129857	0.15901030	0.41091989	1.0000000	0.1857579
JPM	0.3269465	0.27004616	0.30550031	0.34505057	0.17844483	0.1857579	1.0000000
BAC	0.3344743	0.26447502	0.34647709	0.38092750	0.15606849	0.1810116	0.8126908
PG	0.2378719	0.13984608	0.05267596	0.04430315	0.44667362	0.2671080	0.1707378
KO	0.2555036	0.14689499	0.15935765	0.15222502	0.45007317	0.2684131	0.1817924
CAT	0.2565583	0.16672917	0.35173700	0.37645224	0.07414998	0.1457980	0.5101528
	BAC	PG	KO	CAT			
AAPL	0.3344743	0.23787193	0.25550363	0.25655832			
MSFT	0.2644750	0.13984608	0.14689499	0.16672917			
XOM	0.3464771	0.05267596	0.15935765	0.35173700			
CVX	0.3809275	0.04430315	0.15222502	0.37645224			
JNJ	0.1560685	0.44667362	0.45007317	0.07414998			
PFE	0.1810116	0.26710801	0.26841313	0.14579796			
JPM	0.8126908	0.17073784	0.18179235	0.51015281			
BAC	1.0000000	0.17049981	0.17692343	0.49738126			
PG	0.1704998	1.0000000	0.61713490	0.04262906			
KO	0.1769234	0.61713490	1.0000000	0.03653957			
CAT	0.4973813	0.04262906	0.03653957	1.0000000			

```
cor_matrix_B
```

	AAPL	MSFT	NVDA	GOOGL	META	AMZN	AVGO
AAPL	1.0000000	0.5442975	0.4781696	0.5188217	0.4465158	0.5008126	0.4217265
MSFT	0.5442975	1.0000000	0.5662837	0.5593149	0.5081903	0.6115696	0.4543813
NVDA	0.4781696	0.5662837	1.0000000	0.4975046	0.4768501	0.5258929	0.6378619
GOOGL	0.5188217	0.5593149	0.4975046	1.0000000	0.5338640	0.6193969	0.4283939
META	0.4465158	0.5081903	0.4768501	0.5338640	1.0000000	0.5809008	0.3883933
AMZN	0.5008126	0.6115696	0.5258929	0.6193969	0.5809008	1.0000000	0.4360244
AVGO	0.4217265	0.4543813	0.6378619	0.4283939	0.3883933	0.4360244	1.0000000
CRM	0.4037127	0.5406671	0.4269495	0.3918419	0.3876450	0.4662891	0.3198536
ADBE	0.4720552	0.5602995	0.4328442	0.4459463	0.4117990	0.4714146	0.3418862
ORCL	0.2453959	0.4412321	0.4033040	0.2981181	0.2581763	0.3067939	0.4473485
AMD	0.4268173	0.4588499	0.6613708	0.4629016	0.4128703	0.4633692	0.5478882
	CRM	ADBE	ORCL	AMD			
AAPL	0.4037127	0.4720552	0.2453959	0.4268173			
MSFT	0.5406671	0.5602995	0.4412321	0.4588499			
NVDA	0.4269495	0.4328442	0.4033040	0.6613708			
GOOGL	0.3918419	0.4459463	0.2981181	0.4629016			
META	0.3876450	0.4117990	0.2581763	0.4128703			
AMZN	0.4662891	0.4714146	0.3067939	0.4633692			
AVGO	0.3198536	0.3418862	0.4473485	0.5478882			
CRM	1.0000000	0.6378875	0.3067786	0.3029817			
ADBE	0.6378875	1.0000000	0.2957099	0.3182837			
ORCL	0.3067786	0.2957099	1.0000000	0.3303137			
AMD	0.3029817	0.3182837	0.3303137	1.0000000			

```
avg_cor <- function(mat){  
  mean(mat[upper.tri(mat)])  
}  
avg_cor(cor_matrix_A)
```

```
[1] 0.2514609
```

```
avg_cor(cor_matrix_B)
```

```
[1] 0.4515411
```

```
min(cor_matrix_A[upper.tri(cor_matrix_A)])
```

```
[1] 0.01264437
```

```
min(cor_matrix_B[upper.tri(cor_matrix_B)])
```

```
[1] 0.2453959
```

```
max(cor_matrix_A[upper.tri(cor_matrix_A)])
```

```
[1] 0.8553576
```

```
max(cor_matrix_B[upper.tri(cor_matrix_B)])
```

```
[1] 0.6613708
```

```
which(cor_matrix_A == max(cor_matrix_A[upper.tri(cor_matrix_A)]),  
      arr.ind = TRUE)
```

```
      row col  
CVX    4   3  
XOM    3   4
```

```
which(cor_matrix_B == max(cor_matrix_B[upper.tri(cor_matrix_B)]),  
      arr.ind = TRUE)
```

```
      row col  
AMD    11   3  
NVDA   3   11
```

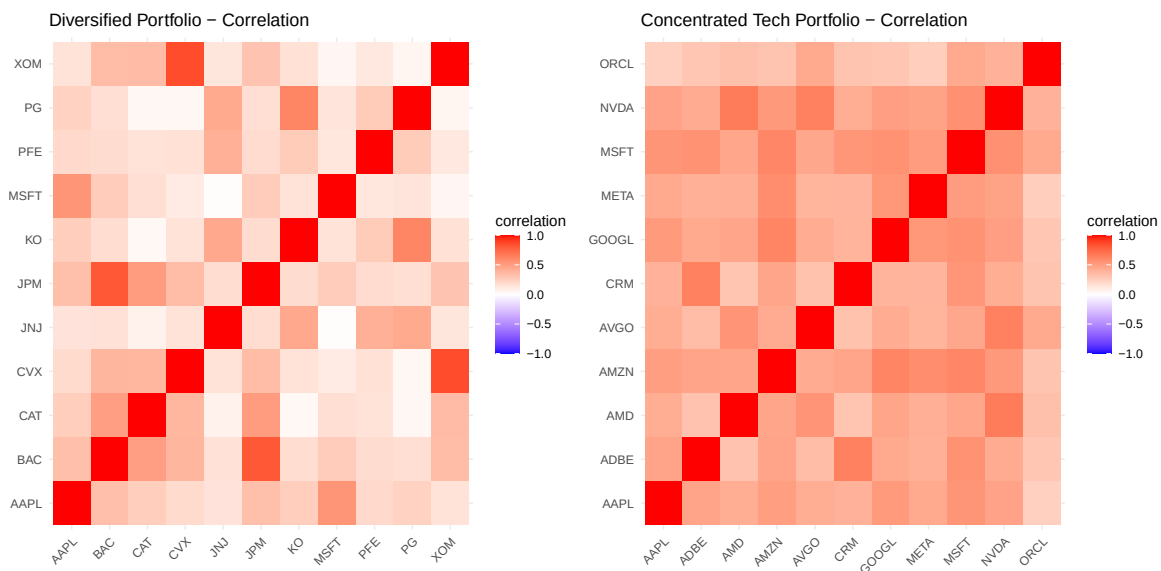
```
plot_corr_heatmap <- function(cor_matrix, title){  
  cor_df <- as.data.frame(cor_matrix) |>  
    mutate(symbol1 = rownames(cor_matrix)) |>  
    pivot_longer(-symbol1, names_to = "symbol2",  
                 values_to = "correlation")  
  ggplot(cor_df, aes(x = symbol1, y = symbol2,  
                     fill = correlation))+
```

```

geom_tile()+
scale_fill_gradient2(low = "blue", mid = "white",
                     high = "red", midpoint = 0,
                     limits = c(-1,1))+
labs(title = title, x = NULL, y = NULL)+
theme_minimal()+
theme(axis.text.x = element_text(angle = 45,
                                   hjust = 1))
}

plot_corr_heatmap(cor_matrix_A,
                  "Diversified Portfolio - Correlation")+
plot_corr_heatmap(cor_matrix_B, "Concentrated Tech Portfolio - Correlation")

```



```
library(MASS, exclude = "select")
```

Attaching package: 'MASS'

The following object is masked from 'package:patchwork':

area

```

set.seed(123)
simulate_portfolio_var <- function(returns_wide, n_sims = 10000,
                                   horizon_days,
                                   confidence_levels = c(0.95, 0.99)){
  returns_matrix <- returns_wide |>
    dplyr::select(-date) |>
    as.matrix()
  n_assets <- ncol(returns_matrix)
  weights <- rep(1/n_assets, n_assets)
  mu <- colMeans(returns_matrix, na.rm = T)
  sigma <- cov(returns_matrix, use = "pairwise.complete.obs")
  simulated_returns <- mvrnorm(n = n_sims, mu = mu, Sigma = sigma)
  simulated_portfolio_returns <- simulated_returns %*% weights
  simulated_portfolio_returns <- simulated_portfolio_returns * sqrt(horizon_days)
  results <- lapply(confidence_levels, function(cl){
    var_threshold <- quantile(simulated_portfolio_returns, probs = 1-cl)
    cvar <- mean(
      simulated_portfolio_returns[simulated_portfolio_returns<=var_threshold])
    tibble(confidence = cl, VaR = -var_threshold, CVaR = -cvar)
  }) |>
    bind_rows()
  list(
    simulated_returns = simulated_portfolio_returns,
    summary = results
  )
}
var_results_A <- simulate_portfolio_var(returns_wide_A,
                                       horizon_days = 10)
var_results_B <- simulate_portfolio_var(returns_wide_B,
                                       horizon_days = 10)
var_results_A$summary

```

```

# A tibble: 2 x 3
  confidence    VaR    CVaR
  <dbl>    <dbl> <dbl>
1     0.95 0.0437 0.0555
2     0.99 0.0626 0.0719

```

```
var_results_B$summary
```

```
# A tibble: 2 x 3
```

	confidence	VaR	CVaR
	<dbl>	<dbl>	<dbl>
1	0.95	0.0866	0.107
2	0.99	0.120	0.139

```
historical_portfolio_returns_A <- returns_wide_A |>
  select(-date) |>
  as.matrix() %*% rep(1/11,11)

realized_10d_A <- zoo::rollapply(
  historical_portfolio_returns_A,
                                width = 10,
                                FUN = sum,
                                align = "right")
var95_A <- var_results_A$summary$VaR[
  var_results_A$summary$confidence == 0.95]
breaches_A <- sum(realized_10d_A < -var95_A)
breaches_rate_A <- breaches_A/length(realized_10d_A)
breaches_rate_A
```

```
[1] 0.03462898
```

```
var_99_A <- var_results_A$summary$VaR[
  var_results_A$summary$confidence == 0.99]

breaches_99_A <- sum(realized_10d_A < -var_99_A)
breach_rate_99_A <- breaches_99_A / length(realized_10d_A)

breach_rate_99_A
```

```
[1] 0.01625442
```

```
historical_portfolio_returns_B <- returns_wide_B |>
  select(-date) |>
  as.matrix() %*% rep(1/11,11)

realized_10d_B <- zoo::rollapply(
  historical_portfolio_returns_B,
                                width = 10,
                                FUN = sum,
                                align = "right")
```

```
var95_B <- var_results_B$summary$VaR[
  var_results_B$summary$confidence == 0.95]
breaches_B <- sum(realized_10d_B < -var95_B)
breaches_rate_B <-breaches_B/length(realized_10d_B)
breaches_rate_B
```

```
[1] 0.05159011
```

```
var_99_B <- var_results_B$summary$VaR[
  var_results_B$summary$confidence == 0.99]

breaches_99_B <- sum(realized_10d_B < -var_99_B)
breach_rate_99_B <- breaches_99_B / length(realized_10d_B)

breach_rate_99_B
```

```
[1] 0.01201413
```

```
library(e1071)
```

Attaching package: 'e1071'

The following objects are masked from 'package:PerformanceAnalytics':

kurtosis, skewness

The following object is masked from 'package:ggplot2':

element

```
kurtosis(historical_portfolio_returns_A)
```

```
[1] 4.999022
```

```
kurtosis(historical_portfolio_returns_B)
```

```
[1] 3.21432
```

```

sim_df <- bind_rows(
  tibble(portfolio = "Diversified", return = as.numeric(
    var_results_A$simulated_returns)),
  tibble(portfolio = "Concentrated Tech", return = as.numeric(
    var_results_B$simulated_returns))
)

var_lines <- bind_rows(
  var_results_A$summary |>
    mutate(portfolio = "Diversified"),
  var_results_B$summary |>
    mutate(portfolio = "Concentrated Tech")
)

ggplot(sim_df, aes(x = return, fill = portfolio))+
  geom_density(alpha = 0.5)+
  geom_vline(data = var_lines |>
    filter(confidence == 0.95), aes(xintercept = -VaR,
                                     color = portfolio),
    linetype = "dashed", linewidth = 0.8)+
  scale_fill_manual(values = c("Diversified" = "steelblue",
                                "Concentrated Tech" = "firebrick"))+
  scale_color_manual(values = c("Diversified" = "steelblue",
                                "Concentrated Tech" = "firebrick"))+
  labs(title = "Simulated 10 day Portfolio Returns",
       subtitle = "Dashed lines mark 95% VaR threshold",
       x = "10 Day Return", y = "Density", fill = "Portfolio")+
  guides(color = "none")+
  theme_minimal()

```

